

APPLYING ADVANCED REGRESSION MODELS AND MACHINE LEARNING
FOR DIFFERENTIATED THYROID CANCER PATIENTS WITH RECURRENCES

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in

Statistical Data Analytics

By

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ABSTRACT

Differentiated Thyroid Cancer Patients with Recurrences

By

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In this study, we aim to predict recurrences of differentiated thyroid cancer. We will use Advanced Regression Models with R programming language and Machine Learning algorithms to find the predicted probability of a patient with differentiated thyroid cancer with recurrences. In addition, we will use a health and medicine dataset such as the Differentiated Thyroid Cancer Recurred from the University of California, Irvine, Machine Learning Repository.

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TABLE OF CONTENTS

Abstract	iv
Acknowledgments.....	v
List of Tables	viii
List of Figures	ix-x
Chapter	
1. Introduction.....	1
2. Advanced Regression Models and Machine Learning.....	5
General and Generalized Linear Regression Model.....	5
Binary Logistic Regression Model.....	6
Probit Model.....	6-7
Complementary Log-Log Model.....	7
Random Forest.....	7-8
3. Differentiated Thyroid Cancer Recurrence Dataset and their results	9
Differentiated Thyroid Cancer Recurrence Dataset.....	9
Exploratory Data Analysis (EDA).....	9-25
Data Cleaning.....	25-30
Results using Generalized Linear Model with a identity link function.....	30-34
Results using Binary Logistic Regression Model with a binomial logit link function.....	34-36
Results using Probit Model with a binomial probit link function.....	36-38
Results using Complementary Log-Log Model with a binomial cloglog link function.....	39-41
Results using Random Forest.....	42-46

4. Conclusion	48
References	49-56

LIST OF TABLES

Table

1. Differentiated Thyroid Cancer Recurrence dataset: Feature Number, Feature Name, Data Type, and Definition of the Feature Name.....12-15
2. Differentiated Thyroid Cancer Recurrence dataset: Feature Name, Instance, and Definition of the Instance Name.....17-24
3. AIC (Akaike Information Criterion), AICc (corrected AIC), and BIC (Bayesian Information Criterion) Values.....41

LIST OF FIGURES

Figure

1. Type of Thyroid Cancer.....	2
2. Anatomy of the Thyroid and Parathyroid Glands.....	3
3. Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 1.....	10
4. Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 2.....	10
5. DTC with Recurrence dataset with num and characters.....	12
6. Differentiated Thyroid Cancer Recurrence dataset in Google Colab in R programming language, part 1.....	16
7. Differentiated Thyroid Cancer Recurrence dataset in Google Colab in R programming language, part 2.....	16
8. Relative Frequency Histogram of Age of a Differentiated Thyroid Cancer with Recurrences Patient Score with Normal Distribution curve.....	25
9. Differentiated Thyroid Cancer with Recurrence dataset with num and factor with number levels	26
10. Differentiated Thyroid Cancer with Recurrence dataset with target variable recurred as num.....	27
11. Summary of Differentiated Thyroid Cancer with Recurrence dataset.....	28
12. Sigmoid Function Applied to the Differentiated Thyroid Cancer Recurrence data with Sigmoid Curve.....	29
13. Missing Data in the Differentiated Thyroid Cancer Recurrence Dataset.....	30
14. Summary of Generalized Linear Model with identity link function.....	31
15. Printing the fitted model to predict for generalized linear model	33

16. Summary of Binary Logistic Regression Model with a binomial logit link function.....	34
17. Printing the fitted model to predict for binary logistic regression.....	36
18. Summary of Probit Model with a binomial logit link function.....	37
19. Printing the fitted model to predict for probit model.....	38
20. Summary of Complementary Log-Log Model with a binomial cloglog link function.....	39
21. Printing the fitted model to predict for Complementary Log-Log model	41
22. Crosstable of the Random Forest.....	43
23. Confusion Matrix of the Random Forest	45
24. Accuracy, Precision, Recall, and F1 Score of the Random Forest	45
25. Accuracy vs. Gini Score of the Random Forest.....	46

CHAPTER 1

Introduction

Thyroid Cancer is a neoplasm of the thyroid gland; a thyroid gland is a highly vascular organ at the neck, usually weighing about 30g, consisting of bilateral lobes connected in the middle by a narrow isthmus [1]. Thyroid cancer is usually characterized by slow growth and a slower and more prolonged clinical course than that of other malignancies [1]. In addition to thyroid cancer, Differentiated Thyroid Cancer (DTC) is a type of thyroid cancer in which the cancer cells look similar to normal thyroid cells when viewed under a microscope, and most differentiated thyroid cancers form in the follicular cells of the thyroid gland [2] that make hormones [3]. A recurrence is the re-appearance of a sign or symptom of a disease after a period of remissions [1]. Recurrence can occur in thyroid cancer.

Differentiated thyroid cancer can be cured on time; this depends on health and medical doctors. Symptoms of thyroid cancer include a lump in your neck, a hoarse voice that lasts for more than 3 weeks, and a sore throat or difficulty swallowing that does not get better [4]. However, it is essential to note that the sooner the health and medical doctors provide a prognosis for the patient, regardless of gender, the better the outcomes and predictions will be now and in the future.

Differentiated Thyroid Cancer (DTC) is split into three types of tumors, that is, PTC-Papillary thyroid cancer (70-80%), FTC-Follicular thyroid cancer (10-15%), and Hurthle cell thyroid cancer (3%), which are the most common which is portrayed in Figure 1 which are the types of thyroid cancer which falls under the Follicular epithelial cells.

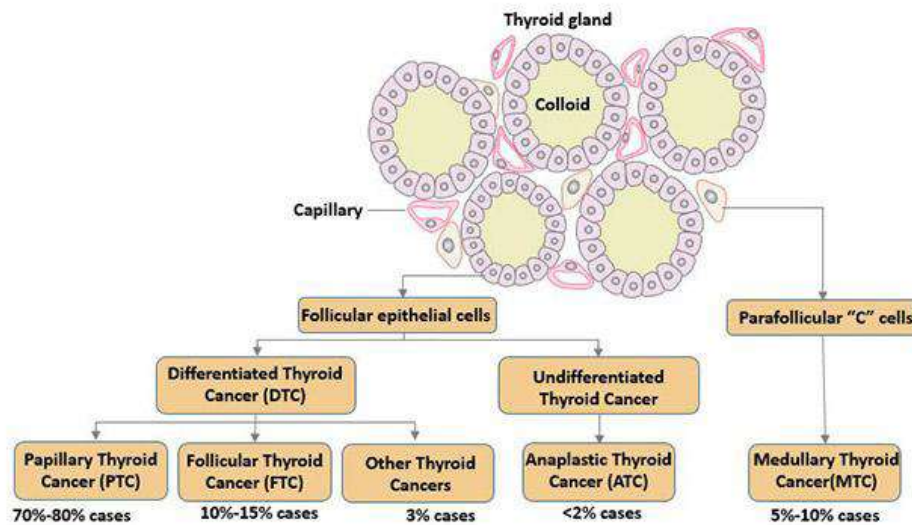


Figure 1. Type of Thyroid Cancer [5].

However, the other type of undifferentiated thyroid cancer also includes the following: anaplastic thyroid cancer. Papillary is the most common type of differentiated thyroid cancer because it accounts for 70-80% of thyroid cancers, spreading to nearby lymph nodes. These types of differentiated thyroid cancer can occur in any gender, male and female, and ages ranging from 30 and up. But it will most likely affect women from ages 50 and up as thyroid cancer becomes more common in middle-aged to older adults than in men. However, it is crucial to note that differentiated thyroid cancer can happen even if the person is a gender of male or female, regardless of their age, and does not smoke or have a history of smoking. The chances are it can still happen to anyone.

An image will be depicted below further to describe a middle-aged female with thyroid and parathyroid glands.

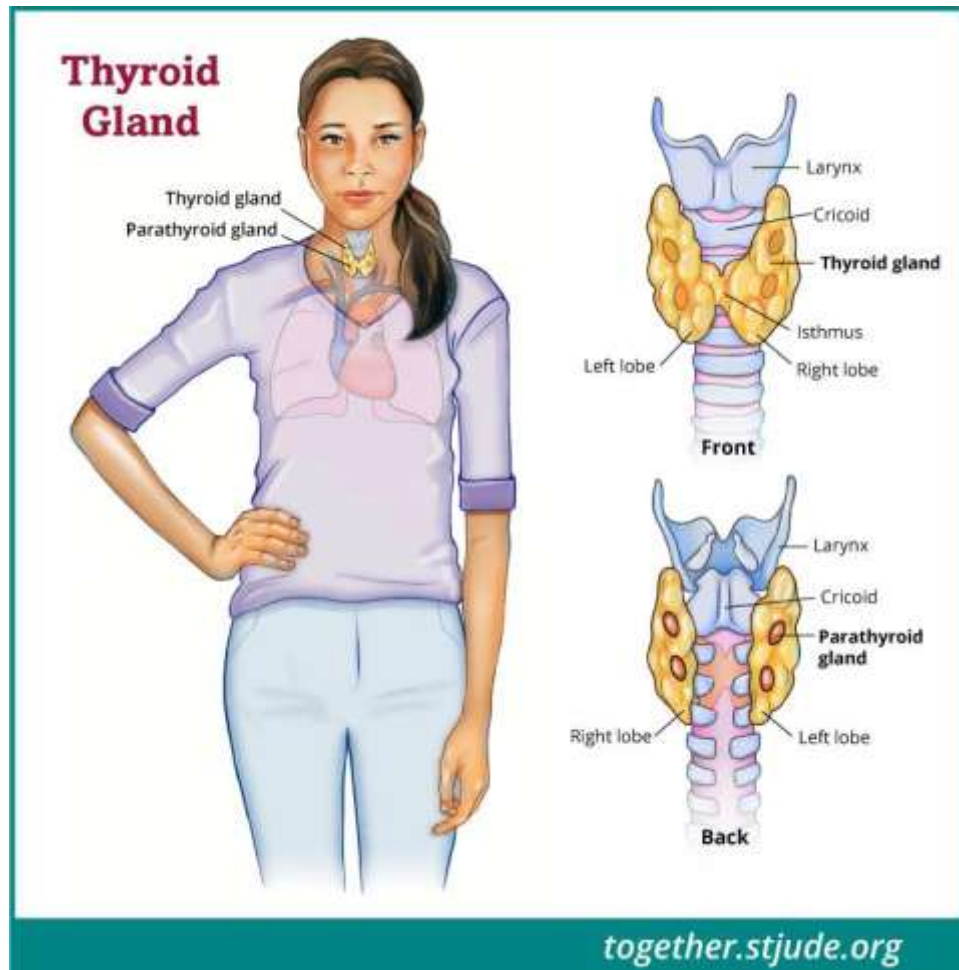


Figure 2. Anatomy of the Thyroid and Parathyroid Glands [3]

Figure 2 shows a middle-aged female with thyroid and parathyroid glands. This Figure 2, shows a thyroid. The thyroid gland lies at the base of the throat in the front of the neck and it has 2 lobes, 1 on the right and the other on the left [3]. Small parathyroid glands lie behind the thyroid [3]. It is essential to note that when a thyroid gland becomes swollen, then there is a chance of the patient having thyroid cancer. This swollen thyroid gland is also known as a ‘goiter.’ Generally, a goiter is a hypertrophic thyroid gland, usually evident as a pronounced swelling in the neck [1]. A goiter can occur in a gland that is producing too much hormone (hyperthyroidism), too little hormone (hypothyroidism), or the correct amount of hormone (euthyroidism) [6]. It also includes

certain carcinogens or substances which are harmful to the throat. Carcin is a combining form meaning 'pertaining to cancer' whereas a carcinogen would be the substance or agent that causes the development or increases the incidence of cancer [1]. These certain carcinogens or substances include smoking, drinking alcohol, drug abuse, are in fact a threat to the throat.

CHAPTER 2

Advanced Regression Models and Machine Learning

Chapter 2 discusses Advanced Regression Models and Machine Learning, their background, the importance of Statistical Learning, and why we apply Advanced Regression to our model as well as the Machine Learning algorithms.

General and Generalized Linear Regression Model

The general linear regression model [7] represents a relation between the response variable y and the predictor variables $x_1, \dots, x_k$ of the form: $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \epsilon$ (1.1) where $\beta_0, \dots, \beta_k$ are constant regression coefficients, and ϵ is a random error that has a normal distribution with the mean zero and a constant variance σ^2 [8]. The parameters of the model $\beta_0, \dots, \beta_k$ and the variance σ^2 are unknown and have to be estimated from the data [8]. In a general linear regression model, the response variable y has a normal distribution with the mean $E(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$ and variance σ^2 [8].

Moreover, in a generalized linear regression model [9], the response variable y is assumed to follow a probability distribution from the exponential family of distributions [8]. In a generalized linear regression model, the regression coefficients $\beta_0, \dots, \beta_k$ and the other parameters of the distribution are estimated by the method of the maximum likelihood estimation [8]. GLM stands for both General and Generalized Linear Regression Model. We find the predictor variable and a response variable using the generalized linear model. Both General and Generalized Linear Regression Model are used in R and Statistical Analysis System (SAS) programming language. Statistical Analysis System (SAS) is statistical software for statical analysis and data analysis.

Binary Logistic Regression Model

Binary Logistic Regression is a statistical analysis model where it uses a binary outcome to predict 1 and 0 and “yes” and “no”. Denote by π the probability that y is equal to one, that is, $\pi = \mathbb{P}(y = 1)$ [8]. Then, $\mathbb{E}(y) = (1)(\pi) + (0)(1 - \pi) = \pi$ [8]. In his 1958 paper "The Regression Analysis of Binary Sequences," D.R. Cox introduced a statistical method to analyze the relationship between one or more independent variables, a sequence of binary values (0 or 1), known as binary logistic regression [10]. The binary (or dichotomous) logistic regression model [10] with the predictors

$$x_1, \dots, x_k \text{ has the form: } \pi = \mathbb{E}(y) = \frac{\exp\{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k\}}{1 + \exp\{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k\}} [8].$$

The name “logistic” comes from the fact that the distribution with the cumulative distribution function $\mathcal{F}(x) = \frac{e^x}{1+e^x}$, $-\infty < x < \infty$, is called the *logistic distribution* and an alternative form of the binary logistic regression model is derived via the use of the logit link function of π (“logit” = “logistic” + “unit”) [8]. We employed the binary logistic regression model because our dataset uses a binary target approach, meaning that our target variables have 'yes' and 'no' with 1 and 0.

Probit Model

Chester Ittner Bliss, introduced the probit model concept, including coining the term "probit" in the 1930's. Consider a binary response variable y and predictors

$x_1, \dots, x_k$ and let $\pi = \mathbb{P}(y = 1) = \mathbb{E}(y)$ [8]. The probit regression model [11]

(“probit” = “probability” + “unit”) is given by: $\pi = \mathbb{E}(y) = \Phi(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)$

where $\Phi(\cdot)$ is the cumulative distribution function of a $\mathcal{N}(0,1)$ distribution [8].

Alternatively, the above relation may be written as $\Phi^{-1}(\pi) = \Phi^{-1}(\mathbb{E}(y)) = \beta_0 +$

$\beta_1 x_1 + \dots + \beta_k x_k$ where the function $\Phi^{-1}(\cdot)$ is called the *probit link function* and that the

probit regression model belongs to the class of generalized linear models [8]. Probit models are good practice because the probit model ensures that the predicted probabilities fall between 0 and 1, as our dataset is binary.

Complementary Log-Log Model

In the 1920's, Sir Ronald Aylmer Fisher introduced the complementary log-log (cloglog) model concept. Consider a binary response variable y and predictors $x_1, \dots, x_k$ and let $\mathbb{P}(y = 1)$, then the *complementary log-log model* [12] has the form $\pi = 1 - \exp\{\exp\{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k\}\}$, or, equivalently, $\ln(-\ln(1-\pi)) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$ [8]. Here the function $\ln(-\ln(1 - \pi))$ is referred to as *complementary log-log link function*, since the log-log function is applied to the complement of π and the function $F(x) = \exp\{-\exp\{-x\}\}$, $-\infty < x < \infty$, that is used in this model, is the cumulative distribution function of the Gumbel or extreme value distribution [8]. Similar to the binary logistic and probit models, the complementary log-log regression represents the class of generalized linear models [8]. Complementary log-log (cloglog) model is good practice because of $\ln$ which represents a small value when dealing with rare events.

Random Forest

Leo Breiman and Adele Cutler developed the Random Forest algorithm [12] in 2001. It is a machine learning technique for classification, prediction, and regression tasks and uses ensemble learning methods. Just like decision trees, the random forest also uses trees. Note that random forest uses an ensemble of trees and lets them vote for the most popular class, and to grow these ensembles, often random vectors are generated that govern the growth of each tree in the ensemble [12]. Random Forest employs a bagging

technique known as bootstrap aggregation where bagging is equal to bootstrap plus aggregation. An early example is bagging (Breiman [1996]), where to grow each tree a random selection (without replacement) is made from the examples in the training set [12]. Usually, it predicts and provides higher accuracy than most machine learning algorithms-models in predictions.

CHAPTER 3

Differentiated Thyroid Cancer Recurrence Dataset and their results

Chapter 3 will discuss the Differentiated Thyroid Cancer Recurrence Dataset and its results and predict the probability of a patient with Thyroid Cancer.

Differentiated Thyroid Cancer Recurrence Dataset

The Differentiated Thyroid Cancer Recurrence dataset is a dataset that was provided by the University of California, Irvine, for health and medicine [13]. Within this dataset, we will use the R programming language to predict the probabilities of Differentiated Thyroid Cancer patients with recurrences. The R programming language is a free software environment for statistical computing and graphics, and it compiles and runs on a wide variety of UNIX platforms, Windows and MacOS [14].

Exploratory Data Analysis (EDA)

To begin with, we first downloaded the Comma-Separated Values (CSV) file from the University of California, Irvine, Machine Learning Repository. Before adding the CSV file to the Google Colab, we first added the CSV file in Microsoft Excel. The CSV file is shown in Microsoft Excel to give a better view of the dataset and what the dataset looks like, as the opinions in Microsoft Excel and Google Colab will vary in differences. The first part of the dataset is shown in Microsoft Excel below. Note that the columns and rows were longer; the figures were cut into two parts. As a result, the two parts, the Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 1, and the Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 2, are shown below in Figures 3 and 4. See below.

	A	B	C	D	E	F	G	H
1	Age	Gender	Smoking	Hx Smoking	Hx Radiotherapy	Thyroid Function	Physical Examination	Adenopathy
2	27	F	No	No	No	Euthyroid	Single nodular goiter-left	No
3	34	F	No	Yes	No	Euthyroid	Multinodular goiter	No
4	30	F	No	No	No	Euthyroid	Single nodular goiter-right	No
5	62	F	No	No	No	Euthyroid	Single nodular goiter-right	No
6	62	F	No	No	No	Euthyroid	Multinodular goiter	No
7	52	M	Yes	No	No	Euthyroid	Multinodular goiter	No
8	41	F	No	Yes	No	Clinical Hyperthyroidism	Single nodular goiter-right	No
9	46	F	No	No	No	Euthyroid	Single nodular goiter-right	No
10	51	F	No	No	No	Euthyroid	Single nodular goiter-right	No
11	40	F	No	No	No	Euthyroid	Single nodular goiter-right	No
12	75	F	No	No	No	Euthyroid	Single nodular goiter-right	No
13	59	F	No	No	No	Euthyroid	Single nodular goiter-left	No
14	49	F	No	No	No	Euthyroid	Multinodular goiter	No
15	50	F	No	No	No	Clinical Hyperthyroidism	Multinodular goiter	No
16	76	F	No	No	No	Clinical Hypothyroidism	Single nodular goiter-right	No
17	42	F	No	No	No	Euthyroid	Single nodular goiter-right	No
18	40	F	No	Yes	No	Euthyroid	Single nodular goiter-right	No
19	44	F	No	No	No	Euthyroid	Multinodular goiter	No
20	43	F	No	No	No	Euthyroid	Multinodular goiter	No
21	52	F	No	No	No	Euthyroid	Multinodular goiter	No
22	41	F	No	No	No	Euthyroid	Single nodular goiter-left	No
23	44	F	No	No	No	Euthyroid	Single nodular goiter-left	No
24	36	F	No	No	No	Euthyroid	Single nodular goiter-right	No
25	70	F	No	No	No	Euthyroid	Single nodular goiter-right	No

Figure 3. Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 1

I	J	K	L	M	N	O	P	Q
Pathology	Focality	Risk	T	N	M	Stage	Response	Recurred
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Indeterminate	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Multi-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Multi-Focal	Low	T1a	NO	MO	I	Indeterminate	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Indeterminate	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Indeterminate	No
Micropapillary	Multi-Focal	Low	T1a	NO	MO	I	Indeterminate	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No
Micropapillary	Uni-Focal	Low	T1a	NO	MO	I	Excellent	No

Figure 4. Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel, part 2

As we saw above in Figures 3 and 4, how a Differentiated Thyroid Cancer Recurrence dataset in Microsoft Excel was shown. Now, we will use a different approach, but in Google Colab, we will use a CSV file using the R programming

language. Likewise, the first way to view the CSV file in Google Colab is through Exploratory Data Analysis (EDA), which is crucial to understanding our dataset. An Exploratory Data Analysis (EDA) is an analysis approach that identifies general patterns in the data, and these patterns include outliers and features of the data that might be unexpected [15].

We will analyze and visualize the datasets in EDA using the Exploratory Data Analysis (EDA) approach. In the R programming language, understand that the figures for the R programming language utilizing Exploratory Data Analysis (EDA) will look different.

Our first step is to start the Google Colab and change the runtime type to R, which is usually done in Runtime on the menu bar, then save and start running our dataset. This is a preeminent and vital step that we must do so that our dataset can run in R because if we do not change our runtime type, we will most likely get an error. After that, we start running in R.

The next approach is to read the .csv file into Colab in R code. Our code would display something like `differentiated_thyroid_cancer_recurrence.data <- read_csv('name/export?format=csv')`. We give our dataset a name and will call this name similar to what our dataset presents. In our code, the assignment operator `<-` assigns a value to a variable [16].

This is the second step in reading the CSV file in Google Colab. After reading the CSV file in Google Colab, we performed Exploratory Data Analysis (EDA) by loading the dataset and importing the libraries needed to run it. This data set contains 13 clinicopathologic features aiming to predict recurrence of well differentiated thyroid

cancer, and was collected in duration of 15 years and each patient was followed for at least 10 years [13].

```
spec_tbl_ [383 × 17] (53: spec_tbl_df/tbl_df/tbl/data.frame)
 $ age          : num [1:383] 27 34 30 62 62 52 41 46 51 40 ...
 $ gender       : chr [1:383] "F" "F" "F" "F" ...
 $ smoking      : chr [1:383] "No" "No" "No" "No" ...
 $ hxsmoking    : chr [1:383] "No" "Yes" "No" "No" ...
 $ hxradiotherapy : chr [1:383] "No" "No" "No" "No" ...
 $ thyroidfunction : chr [1:383] "Euthyroid" "Euthyroid" "Euthyroid" "Euthyroid" ...
 $ physical examination: chr [1:383] "Single nodular goiter-left" "Multinodular goiter" "Single nodular goiter-right"
 $ adenopathy    : chr [1:383] "No" "No" "No" "No" ...
 $ pathology     : chr [1:383] "Micropapillary" "Micropapillary" "Micropapillary" "Micropapillary" ...
 $ focality      : chr [1:383] "Uni-Focal" "Uni-Focal" "Uni-Focal" "Uni-Focal" ...
 $ risk          : chr [1:383] "Low" "Low" "Low" "Low" ...
 $ t            : chr [1:383] "T1a" "T1a" "T1a" "T1a" ...
 $ n            : chr [1:383] "N0" "N0" "N0" "N0" ...
 $ m            : chr [1:383] "M0" "M0" "M0" "M0" ...
 $ stage        : chr [1:383] "I" "I" "I" "I" ...
 $ response      : chr [1:383] "Indeterminate" "Excellent" "Excellent" "Excellent" ...
 $ recurred     : chr [1:383] "No" "No" "No" "No" ...
```

Figure 5. DTC with Recurrence dataset with num and characters

In Figure 5, a DTC with Recurrence dataset with num and characters is shown above. Within our dataset are 383 instances (rows) and 17 features (columns), where 16 columns represent a character, one column represents a double, and our last column is the target column. This is the beginning where the dataset is shown in EDA. Sixteen features (columns) represent character meaning that some data cleaning needs to be done to ensure that the dataset runs correctly. A table with their feature numbers, names, data types, and definition of the feature name will be shown in Table 1.

Table 1

Differentiated Thyroid Cancer Recurrence dataset: Feature Number, Feature Name,

Data Type, and Definition of the Feature Name

Feature Number	Feature Name	Data Type	Definition

0	Age	Double	The period of time someone has been alive or something has existed [17].
1	Gender	Character	A group of people in a society who share particular qualities or ways of behaving which that society associates with being male, female, or another identity [18].
2	Smoking	Character	The action of smoking a cigarette, pipe, etc., or the activity of doing this regularly [19].
3	Hx smoking	Character	Hx stands for History of Smoking. The patient's past and current use of tobacco and forms and types of tobacco product use.
4	Hx radiotherapy	Character	Hx stands for History of Radio Therapy. Radiotherapy means the use of radiation, usually x-rays, to treat cancer [20].
5	Thyroid Function	Character	The process by which the thyroid gland produces and releases hormones that controls your body's metabolism, the process in which

			your body transforms the food you eat into energy [21]. Thyroid hormone actually represents the combination of the two main hormones that your thyroid gland releases: thyroxine (T4) and triiodothyronine (T3) [21]. They're often collectively referred to as "thyroid hormone" because T4 is largely inactive, meaning it doesn't impact your cells, whereas T3 is active [21]. Once your thyroid releases T4, certain organs in your body transform it into T3 so that it can impact your cells and your metabolism [21].
6	Physical Examination	Character	An examination of a person's body by a doctor in order to discover if that person is healthy [22].
7	Adenopathy	Character	An enlargement of any gland [1].
8	Pathology	Character	The scientific study of disease [23].
9	Focality	Character	Pertaining to a focus [1].

10	Risk	Character	The possibility of something bad happening [24].
11	T	Character	Abbreviation for tumor [1], [25]. This describes the primary tumor size [26].
12	N	Character	Abbreviation for nearby lymph nodes [25]. This indicates whether the thyroid cancer cells have spread to regional lymph nodes [26].
13	M	Character	Abbreviation for metastasis [1], [25]. This refers to whether the cancer has metastasized (spread to distant areas of the body) [26].
14	Stage	Character	A part of an activity or a period of development [27].
15	Response	Character	An answer or reaction [28].
16	Recurred	Character	A disease sign or symptom that returns periodically [1].

In the overhead, we showed Table 1 with the Differentiated Thyroid Cancer Recurrence dataset: Feature Number, Feature Name, Data Type, and Definition of the Feature Name. We will next exhibit how reading the data frame with ‘read_csv’ displays in Google Colab. By default, it shows the first six rows before displaying Figure 6; a library (reader) was used to import the Differentiated Thyroid Cancer Recurrence dataset into R with read.

age	gender	smoking	hxsmoking	hxradiotherapy	thyroidfunction	physicalexamination
<dbl>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
27	F	No	No	No	Euthyroid	Single nodular goiter-left
34	F	No	Yes	No	Euthyroid	Multinodular goiter
30	F	No	No	No	Euthyroid	Single nodular goiter-right
62	F	No	No	No	Euthyroid	Single nodular goiter-right
62	F	No	No	No	Euthyroid	Multinodular goiter
52	M	Yes	No	No	Euthyroid	Multinodular goiter

Figure 6. Differentiated Thyroid Cancer Recurrence dataset in Google Colab in R

programming language, part 1

adenopathy	pathology	focality	risk	t	n	m	stage	response	recurred
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Indeterminate	No
No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
No	Micropapillary	Multi-Focal	Low	T1a	N0	M0	I	Excellent	No
No	Micropapillary	Multi-Focal	Low	T1a	N0	M0	I	Indeterminate	No

Figure 7. Differentiated Thyroid Cancer Recurrence dataset in Google Colab in R programming language, part 2

As shown above, Figures 6 and 7 are shown. These figures show the Differentiated Thyroid Cancer Recurrence dataset in Google Colab in R programming

language. The figures practically resemble those in the Microsoft Excel Comma-Separated Values (CSV) file. However, when we print the data, it is entirely diverse.

Table 2

Differentiated Thyroid Cancer Recurrence dataset: Feature Name, Instance, and Definition of the Instance Name

Feature Name	Instance	Definition
Thyroid Function	Clinical Hyperthyroidism	An overactive thyroid. A condition characterized by hyperactivity of the thyroid gland [1]. The gland is usually enlarged, secreting greater than normal amounts of thyroid hormones, and the metabolic processes of the body are accelerated [1].
Thyroid Function	Euthyroid	Pertaining to a normal thyroid gland [1].
Thyroid Function	Subclinical Hypothyroidism	Subclinical hyperthyroidism is defined as a state of suppression of serum TSH with normal free thyroxine and triiodothyronine levels in a patient who lacks clinical features of thyrotoxicosis [29].
Physical Examination	Diffuse goiter	Detecting an enlargement of all parts of the thyroid gland [1].
Physical Examination	Multinodular goiter	Multinodular goiters can be either a toxic multinodular goiter (i.e. makes too much

		thyroid hormone and causes hyperthyroidism [30].
Physical Examination	Normal	Describing a standard, average, or typical example of a set of objects or values [1].
Physical Examination	Single nodular goiter-left	A single enlarged goiter that contains nodules [1] on the left side of the throat.
Physical Examination	Single nodular goiter-right	A single enlarged goiter that contains nodules [1] on the right side of the throat.
Adenopathy	Bilateral	Occurs on both sides of the throat [31].
Adenopathy	Extensive	Extensive adenopathy, also known as generalized lymphadenopathy, can be defined as enlargement of more than two non-contiguous lymph node groups [32].
Adenopathy	Left	Occurs on left side of the throat [31].
Adenopathy	No	No enlargement of any glands [1].
Adenopathy	Posterior	Occurs on back side of the throat [1], [31].
Adenopathy	Right	Occurs on right side of the throat [31].
Pathology	Follicular	Follicular thyroid cancer, or follicular carcinoma, is the second most common type of thyroid cancer [26]. It is found more

		frequently in countries with an inadequate dietary intake of iodine [26].
Pathology	Hurthle cell	A benign tumor of the thyroid gland composed of large cells with granular eosinophilic cytoplasm [1]. A malignant neoplasm of the thyroid gland composed of Hurtle cells [1]. Hürthle cell thyroid cancer, also known as Hürthle cell carcinoma and oxyphil cell carcinoma, is a rare thyroid cancer [26].
Pathology	Micropapillary	Micropapillary thyroid cancer, also known as papillary thyroid microcarcinoma (PTMC), is a type of papillary thyroid cancer (PTC) specifically defined where microcarcinomas are thyroid cancers < 1 cm in size [33].
Pathology	Papillary	Of or pertaining to a papilla [1]. Papillary thyroid cancer, or papillary carcinoma, is the most common type of thyroid cancer [26]. Papillary carcinomas are slow-growing, differentiated cancers that develop from follicular cells and in one or both lobes of the thyroid gland [26].

Focality	Multi-Focal	The simultaneous presence of more than 1 tumor focus within the thyroid gland [34].
Focality	Uni-Focal	Unifocal refers to the primary tumour having a single foci of cancer cells [35] within the thyroid gland.
Risk	High	The cancer has grown through the thyroid and into many tissues in the neck (called gross extension) and has spread to other parts of the body (called distant metastases) [36].
Risk	Intermediate	Has spread to more than 5 lymph nodes in the neck, invaded blood vessels (called vascular invasion), and grown through the thyroid and into tissues around the thyroid [36]
Risk	Low	Not grown through the thyroid, has not spread to nearby tissues or other parts of the body, and is 4 cm or smaller [36].
T	T1a	T1 means the cancer is within the thyroid gland and is 2cm or less [37]. T1a means the cancer is completely inside the thyroid. It is 1cm across or smaller [37].
T	T1b	T1b means the cancer is completely inside the thyroid [37]. It is between 1cm and 2cm across [37].

T	T2	T2 means the cancer is completely inside the thyroid [37]. It is more than 2cm but no greater than 4cm across [37].
T	T3a	T3a means the cancer is more than 4cm across [37]. It is still completely inside the thyroid [37].
T	T3b	T3b means the cancer is any size [37]. It has grown outside the thyroid into one or more of the muscles beside the thyroid (strap muscles) [37].
T	T4a	T4 cancers have grown outside the thyroid [37]. T4a means the cancer has grown outside the thyroid into nearby soft tissue, such as the voice box (larynx), windpipe (trachea), food pipe (oesophagus) or the voice box nerve (recurrent laryngeal nerve) [37].
T	T4b	T4b means the cancer has grown outside the thyroid into the area surrounding the bones of the spine or one of the main blood vessels in the neck area [37].
N	N0	N describes whether the cancer has spread to the lymph nodes [37]. N0 means that the

		cancer hasn't spread to nearby lymph nodes [37].
N	N1a	N1 means the cancer has spread to nearby lymph nodes [37]. N1a means the cancer has spread to lymph nodes close to the thyroid in the neck [37].
N	N1b	N1b means the cancer has spread to other lymph nodes in the neck [37]. These are called cervical lymph nodes [37]. Or it has spread to lymph nodes behind the throat, called the retropharyngeal lymph nodes [37].
M	M0	M0 means the cancer hasn't spread to another part of the body [37].
M	M1	M1 means the cancer has spread to another part of the body, such as the lungs or bones [37].
Stage	I	The primary tumor can be any size and the cancer may or may not have spread to lymph nodes for patients under the age of 55 [26]. The tumor is 2 centimeters (cm) or smaller (less than an inch wide), and has not grown outside the thyroid. It has not spread to nearby lymph nodes or distant sites [26].

Stage	II	The primary tumor can be any size and the cancer may or may not have spread to lymph nodes, but cancer cells have spread to distant areas of the body for patients under the age of 55 [26]. The diameter of the primary tumor ranges from 2 to 4 cm [26]. No cancer cells are in regional lymph nodes or distant sites in the body [26].
Stage	III	The primary tumor is larger than 4 cm, or has grown outside the thyroid, but has not spread to nearby lymph nodes or beyond (differentiated cancers only) [26].
Stage	IVA	Cancers at this stage have grown beyond the thyroid gland and may have spread into nearby tissue, or they may have spread to lymph nodes in the neck and upper chest, but not to distant sites [26].
Stage	IVB	The primary tumor has grown into the spine or into nearby large blood vessels. In this thyroid cancer stage, the disease may or may not have spread to lymph nodes, but has not reached distant sites [26].

Response	Biochemical Incomplete	Biochemical incomplete response (BIR) is defined as elevated thyroglobulin or rising thyroglobulin antibodies following treatment without structural evidence of disease at 1–2 years after initial treatment [38].
Response	Excellent	No clinical, biochemical, or structural evidence of disease is present after treatment [39].
Response	Indeterminate	Non-specific biochemical or structural findings which cannot be confidently classified as either benign or malignant [39].
Response	Structural Incomplete	Indicates persistent or newly identified local or distant metastases, even after initial treatment, regardless of the levels of thyroglobulin (Tg) or anti-Tg antibodies [39].

Table 2 shows the Differentiated Thyroid Cancer Recurrence dataset with its feature name, instance, and the definition of the instance name. Finally, with the definitions of our feature names, we understood the dataset more strategically. In other words, it is vital to understand what your dataset means when we start to program in R.

We attach our file after understanding our dataset, target variable, and feature names. The `attach()` function makes the variables within a data frame directly accessible in our R environment. We used the code

`attach(differentiated_thyroid_cancer_recurrence.data)` to do this in R. In this code, we can see the name attached in the front of the code, and it attaches the file. By attaching the file in the R environment, we must ensure that the dataset has no missing values and do data cleaning. Before we do data cleaning, a histogram of age will be displayed.

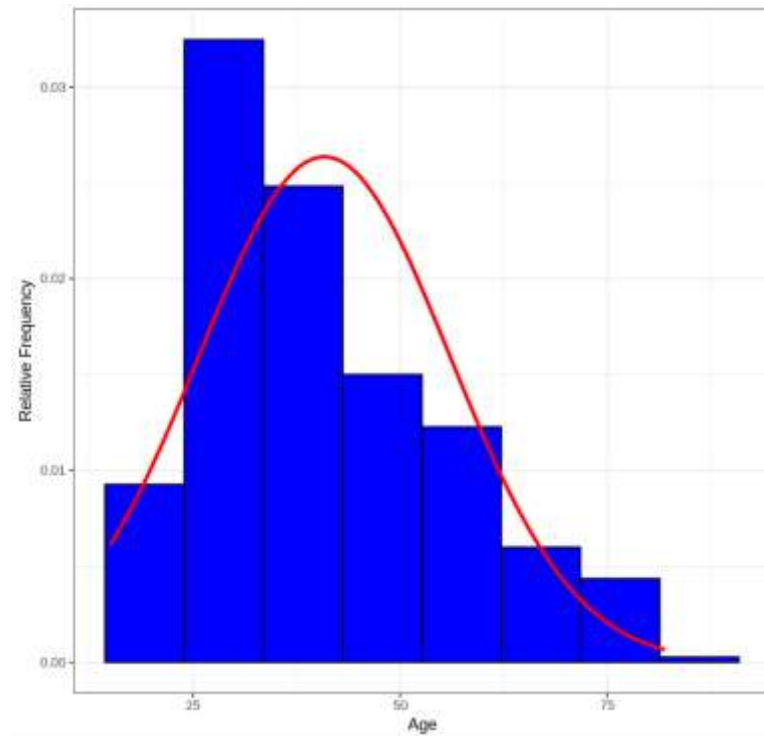


Figure 8. Relative Frequency Histogram of Age of a Differentiated Thyroid Cancer with Recurrences Patient Score with Normal Distribution curve

Figure 8 shows how the histogram looks in R because age consists of a number and not a character. In this Relative Frequency Histogram of age, no data cleaning was performed. See above.

Data Cleaning

Data cleaning is performed to remove inaccurate data, clean the data, and correct errors. In the differentiated thyroid cancer recurrence data, 19 duplicated rows were found using the `duplicated()` method. The `na.omit()` function was used to handle missing

values in the character's objects, and the `is.na()` function in R. To check if the sum of duplicated rows was duplicated, `sum(duplicated())` in R was used, and our output was 0, according to our code, which means that everything -- that our data is cleaned. Our dataset still shows characters, so we need to convert the character columns to factors. First, we install the ordinal package before loading the library. Then, we convert the character columns to factors. The converted character columns to factors are shown below.

```
tibble [364 × 17] (53: tbl_df/tbl/data.frame)
 $ age      : num [1:364] 27 34 30 62 62 52 41 46 51 40 ...
 $ gender    : chr [1:364] "F" "F" "F" "F" ...
 $ smoking   : chr [1:364] "No" "No" "No" "No" ...
 $ hxsmoking : chr [1:364] "No" "Yes" "No" "No" ...
 $ hxradioterapy : chr [1:364] "No" "No" "No" "No" ...
 $ thyroidfunction : chr [1:364] "Euthyroid" "Euthyroid" "Euthyroid" "Euthyroid" ...
 $ physicalexamination: chr [1:364] "Single nodular goiter-left" "Multinodular goiter" "Single nodular goiter-right"
 $ adenopathy : chr [1:364] "No" "No" "No" "No" ...
 $ pathology  : chr [1:364] "Micropapillary" "Micropapillary" "Micropapillary" "Micropapillary" ...
 $ focality   : chr [1:364] "Uni-Focal" "Uni-Focal" "Uni-Focal" "Uni-Focal" ...
 $ risk       : chr [1:364] "Low" "Low" "Low" "Low" ...
 $ t          : chr [1:364] "T1a" "T1a" "T1a" "T1a" ...
 $ n          : chr [1:364] "N0" "N0" "N0" "N0" ...
 $ m          : chr [1:364] "M0" "M0" "M0" "M0" ...
 $ stage      : chr [1:364] "I" "I" "I" "I" ...
 $ response   : chr [1:364] "Indeterminate" "Excellent" "Excellent" "Excellent" ...
 $ recurred   : chr [1:364] "No" "No" "No" "No" ...
```

Figure 9. Differentiated Thyroid Cancer with Recurrence dataset with num and factor with number levels

In Figure 9, the characters are now converted to factors with number levels ranging from 2 to 7 levels. Within these ranges, the names of the factors with number levels are shown. For instance, in gender, there are factors with two levels where the two levels would mean Female for one and Male for two. There is a total of 364 observations and 17 variables this includes removing the duplicates. However, our target variable “recurred” shows a character, which in fact needs to be changed to num and not char. Since our target variable recurred shows No-Yes according to our data, then this means it is a binary approach.

When the columns are not factored, we will run into an error. An `as.factor()` function in R was employed to convert a vector object to a factor which is the correct approach. Another approach can be when we assign a new variable name to our data frame and we want to call this `recurred_new` because adding a new variable name to our data frame will make work if we wanted to derive from the name 'recurred'. This approach will work, but, however, it might not be good practice. Then, we run an if-else statement in R. An `ifelse` returns a value with the same shape as `test` which is filled with elements selected from either yes or no depending on whether the element of `test` is TRUE or FALSE [40]. In `ifelse` statement, 1 means Yes, and 0 means No. After running `ifelse`, we implement `recurred <- as.numeric(differentiated_thyroid_cancer_recurrence.data$recurred)`. Our code utilizes `as.numeric()` because we need to convert it into numerical values. Following the above, the next code becomes: `unique(differentiated_thyroid_cancer_recurrence.data$recurred)` because we need to check for binary digits and this becomes 0 and 1.

```
tibble [364 × 17] (S3: tbl_df/tbl/data.frame)
 $ age          : num [1:364] 27 34 30 62 62 52 41 46 51 40 ...
 $ gender       : Factor w/ 2 levels "F","M": 1 1 1 1 1 2 1 1 1 1 ...
 $ smoking      : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 1 1 1 1 ...
 $ hxsmoking    : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 2 1 1 1 ...
 $ hxradiotherapy : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ thyroidfunction : Factor w/ 5 levels "Clinical Hyperthyroidism",...: 3 3 3 3 3 3 1 3 3 3 ...
 $ physical examination: Factor w/ 5 levels "Diffuse goiter",...: 4 2 5 5 2 2 5 5 5 5 ...
 $ adenopathy    : Factor w/ 6 levels "Bilateral","Extensive",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ pathology     : Factor w/ 4 levels "Follicular","Hurthel cell",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ focality      : Factor w/ 2 levels "Multi-Focal",...: 2 2 2 2 1 1 2 2 2 2 ...
 $ risk          : Factor w/ 3 levels "High","Intermediate",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ t             : Factor w/ 7 levels "T1a","T1b","T2",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ n             : Factor w/ 3 levels "N0","N1a","N1b": 1 1 1 1 1 1 1 1 1 1 ...
 $ m             : Factor w/ 2 levels "M0","M1": 1 1 1 1 1 1 1 1 1 1 ...
 $ stage         : Factor w/ 5 levels "I","II","III",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ response      : Factor w/ 4 levels "Biochemical Incomplete",...: 3 2 2 2 2 3 2 2 2 2 ...
 $ recurred      : num [1:364] 0 0 0 0 0 0 0 0 0 0 ...
```

Figure 10. Differentiated Thyroid Cancer with Recurrence dataset with target variable `recurred` as num

In Figure 10, a figure will be shown to show the new data.frame. Hence, we see num now in the data.frame meaning that we have created a new variable, and it is good practice. After deriving the code, we then check for our target variable that we assigned to our data frame. Above in Figure 10, exhibits a complete output of the dataframe where the last row shows a binary num. We have converted recurred to numeric with 1 being 'Yes' and 0 being 'No'. Additionally, a summary is shown above in Figure 10 to show the summaries of all Factor w/ 2, 3, 4, 5, 6, and 7 levels. This is because each Factor w/ any number levels show the summaries involved with certain levels. These levels are based on Tables 1 and 2. Furthermore, the definitions of some of these levels are shown in Table 2. Therefore, this summary portrays an important methodology in understanding our dataset with where we not only performed EDA on the dataset but data cleaning as well. This summary will be shown in Figure 11, see below.

```

      age      gender  smoking  hxsmoking  hxradiotherapy
Min.   :15.00  F:293  No :315  No :336  No :357
1st Qu.:38.00  M: 71  Yes: 49  Yes: 28  Yes:  7
Median :38.00
Mean   :41.25
3rd Qu.:52.00
Max.   :82.00

      thyroidfunction      physicalexamination
Clinical Hyperthyroidism : 20  Diffuse goiter      : 7
Clinical Hypothyroidism  : 12  Multinodular goiter :135
Euthyroid                 :313  Normal             : 7
Subclinical Hyperthyroidism: 5   Single nodular goiter-left : 88
Subclinical Hypothyroidism : 14  Single nodular goiter-right:127

      adenopathy      pathology      focality      risk
Bilateral: 32  Follicular      : 28  Multi-Focal:136  High      : 32
Extensive:  7  Hurthel cell   : 20  Uni-Focal  :228  Intermediate:102
Left       : 17  Micropapillary: 45                      Low       :230
No         :258  Papillary      :271
Posterior:  2
Right      : 48

      t      n      m      stage      response      recurred
T1a: 46  N0 :249  M0:346  I   :314  Biochemical Incomplete: 23  No :256
T1b: 40  N1a: 22  M1: 18  II  : 32  Excellent             :189  Yes:108
T2 :138  N1b: 93                      III:  4  Indeterminate         : 61
T3a: 96                      IVA:  3  Structural Incomplete : 91
T3b: 16                      IVB: 11
T4a: 20
T4b:  8

```

Figure 11. Summary of Differentiated Thyroid Cancer with Recurrence dataset

To check for the recurred in binary, we will use a library(ggplot2) that had been installed properly in Google Colab to ensure that we can plot the target variable. Whereas showing in ggplot2 a plot that shows only yes and no, shown below in Figure 12.

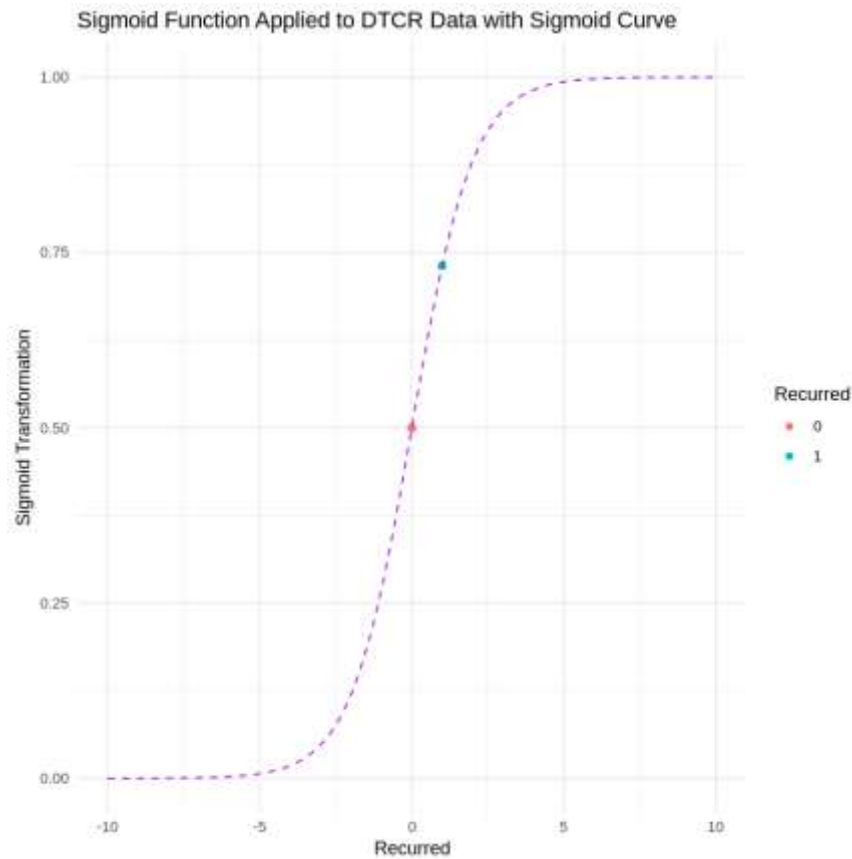


Figure 12. Sigmoid Function Applied to the Differentiated Thyroid Cancer Recurrence data with Sigmoid Curve

Figure 12 exhibits the Sigmoid Function Applied to the Differentiated Thyroid Cancer Recurrence data with Sigmoid Curve. According to our dataset, recurred has two levels now; yes and no and these numeric are numbers and not characters as we did not want to derive from the name 'recurred' and leave as is.

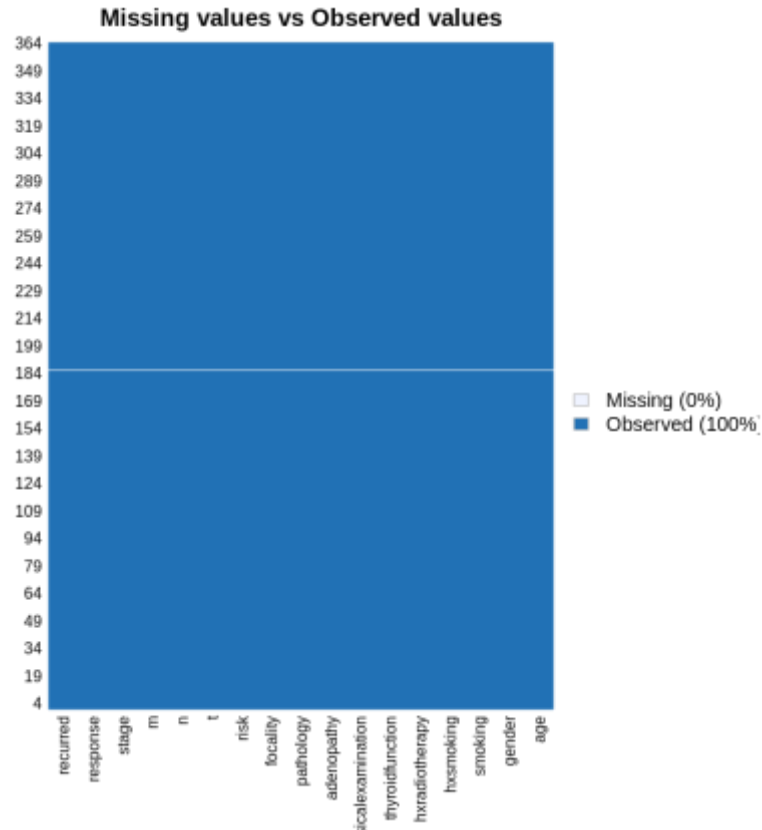


Figure 13. Missing Data in the Differentiated Thyroid Cancer Recurrence Dataset

Lastly, we will use the Amelia package to study any missing values in the graph before running the Generalized Linear Model and Machine Learning Model in R. To do this, the "Amelia" package must be installed first with the library. After installing this package, we ran the R programming language code. Figure 13 shows missing values vs observed values to glimpse if any missing values need to be examined.

Results using Generalized Linear Model with a identity link function

Chapter 2 noted what the Generalized Linear Regression Model demonstrates, and now we will use the GLM() for the Gaussian Family with identity link function. Before we use the Generalized Linear Model with the identity link function, we need to create indicator variables, which are also called dummy variables, because we want to use only those critical to our regression model. Therefore, these variables are categorical

variables and not numeric. For example, a categorical variable might look like something where a gender has “M” for Male and “F” for Female. This is an example of a categorical variable. By creating indicator variables to characterize the categorical variables in regression models, we likewise need to relevel the categorical variables, which is purely theoretical. In this, we fit the generalized linear model and then print a summary of our model, which is shown in Figure 14.

```
glm(formula = recurred ~ age + gender.rel + hasmoking.rel + thyroidfunction.rel +
      t.rel + n.rel + m.rel + stage.rel, family = gaussian(link = identity),
      data = differentiated_thyroid_cancer_recurrence.data)
```

Coefficients:	Estimate	Std. Error	t value
(Intercept)	0.366567	0.208763	1.269
age	0.001858	0.001184	1.337
gender.relF	-0.341126	0.044320	-3.191
hasmoking.relYes	0.040116	0.066156	0.606
thyroidfunction.relClinical Hyperthyroidism	-0.045039	0.156943	-0.287
thyroidfunction.relClinical Hypothyroidism	0.076381	0.166945	0.458
thyroidfunction.relfuthyroid	0.076396	0.142237	0.537
thyroidfunction.relSubclinical Hypothyroidism	-0.152601	0.167460	-0.911
t.relT1b	0.084858	0.068072	1.247
t.relT2	0.034236	0.053839	0.636
t.relT3a	0.181732	0.059046	3.078
t.relT3b	0.482372	0.098735	4.886
t.relT4a	0.406101	0.102551	3.968
t.relT4b	0.056981	0.161002	0.354
n.relN0	-0.222992	0.073903	-3.017
n.relN1b	0.241382	0.076833	3.142
m.relM0	-0.086819	0.118926	-0.730
stage.relI	-0.064402	0.189350	-0.340
stage.relIII	0.104550	0.185950	0.562
stage.relIVA	0.414164	0.257835	1.606
stage.relIVB	0.192349	0.223712	0.860

	Pr(> t)
(Intercept)	0.20514
age	0.18223
gender.relF	0.00155 **
hasmoking.relYes	0.54465
thyroidfunction.relClinical Hyperthyroidism	0.77430
thyroidfunction.relClinical Hypothyroidism	0.64759
thyroidfunction.relfuthyroid	0.59154
thyroidfunction.relSubclinical Hypothyroidism	0.36280
t.relT1b	0.21540
t.relT2	0.52512
t.relT3a	0.00225 **
t.relT3b	1.59e-06 ***
t.relT4a	9.12e-05 ***
t.relT4b	0.72381
n.relN0	0.00274 **
n.relN1b	0.00183 **
m.relM0	0.46588
stage.relI	0.73397
stage.relIII	0.57431
stage.relIVA	0.10912
stage.relIVB	0.39050

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

[Dispersion parameter for gaussian family taken to be 0.89456176]

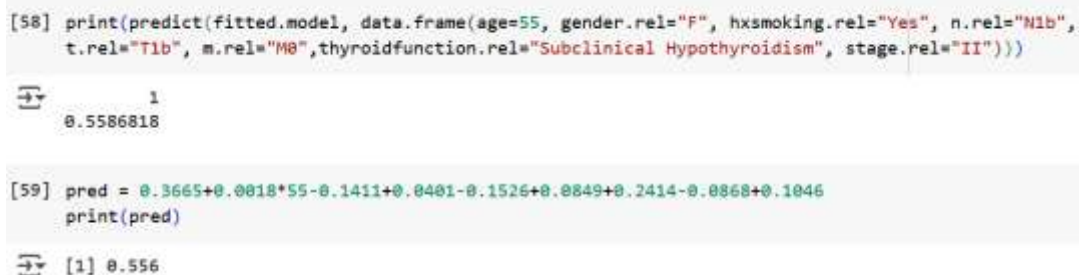
Null deviance: 75.956 on 363 degrees of freedom
 Residual deviance: 32.435 on 343 degrees of freedom
 AIC: 196.86

Number of Fisher Scoring iterations: 2
 0.307508964036492

Figure 14. Summary of Generalized Linear Model with identity link function

Figure 14 shows a summary of the Generalized Linear Model with an identity link function. Now that we have fitted the regression model, we will specify all the estimated parameters. From this output, the fitted regression model is: $\widehat{E}(\text{score}) = 0.3665 + 0.0018 \cdot \text{Age} - 0.1411 \cdot \text{Female} - 0.0401 \cdot \text{No History of Smoking} - 0.0450 \cdot \text{Clinical Hyperthyroidism} + 0.0764 \cdot \text{Clinical Hyperthyroidism} + 0.0764 \cdot \text{Euthyroid} - 0.1526 \cdot \text{Subclinical Hypothyroidism} + 0.0849 \cdot \text{T1b} + 0.0342 \cdot \text{T2} + 0.1817 \cdot \text{T3a} + 0.4824 \cdot \text{T3b} + 0.4061 \cdot \text{T4a} + 0.0569 \cdot \text{T4b} - 0.2230 \cdot \text{N0} + 0.2414 \cdot \text{N1b} - 0.0868 \cdot \text{M0} - 0.0644 \cdot \text{Stage I} + 0.1046 \cdot \text{Stage II} + 0.4142 \cdot \text{Stage IVA} + 0.1923 \cdot \text{Stage IVB}$. The estimated standard deviation of the error is $\hat{\sigma} = 0.3075$. According to our data, gender, T3a, T3b, T4a, N0, N1b, however, are statistically significant predictors of differentiated thyroid cancer recurrence score at the 5% level of significance, since the corresponding values are less than 0.05. The estimated significant regression coefficients are gender, T3a, T3b, T4a, N0, and N1b. The estimated mean satisfaction score for Female Gender is 0.1411 points smaller than that of Male Gender for a recurrence in patients with different thyroid cancers. In addition, with a T level decrease in recurrence in patients with different thyroid cancer, the estimated average is increased by 0.1817 for T3a level, 0.4824 for T3b level, and 0.4061 for T4a level. The estimated mean satisfaction score for a recurrence in patients with different thyroid cancer is 0.2230, decreased with a N N0 level, than that of an increased recurrence in patients with different thyroid cancer, and 0.2414, increased with a N N1b level, then that of a decreased recurrence in patients with different thyroid cancer. Log-likelihood is a measure of the goodness of fit of a statistical model in regression analysis. We note from the output that the log-likelihood function for

the fitted model equals -76.4312. Next, we fit the intercept-only model, and now we can calculate the deviance statistic and carry out the goodness-of-fit test. Since the p-value (8.26313e-54) is small and not more than 0.05, we conclude that the fitted model fits better than the intercept-only model. We use the fitted model to predict our outcome for the regression model. Lastly, suppose we would like to use the fitted model to predict the history of smoking score for a recurrence in patients with differentiated thyroid cancer where the patient is a 55-year-old female who has a history of smoking, with N1b, T1b, M0, a subclinical hypothyroidism, and is in stage II. We compute using $Hxsmoker^0 = 0.3665 + 0.0018 \cdot Age\ 55 - 0.1411 \cdot Female + 0.0401\ Yes\ History\ of\ Smoking - 0.1526 \cdot Subclinical\ Hypothyroidism + 0.0849 \cdot T1b + 0.2414 \cdot N1b - 0.0868 \cdot M0 + 0.1046 \cdot Stage\ II = 0.556$ which is approximate to our answer being 0.5586818 and is shown in Figure 15.



```
[58] print(predict(fitted.model, data.frame(age=55, gender.rel="F", hxsmoking.rel="Yes", n.rel="N1b",
      t.rel="T1b", m.rel="M0", thyroidfunction.rel="Subclinical Hypothyroidism", stage.rel="II")))
      1
      0.5586818

[59] pred = 0.3665+0.0018*55-0.1411+0.0401-0.1526+0.0849+0.2414-0.0868+0.1046
      print(pred)
      [1] 0.556
```

Figure 15. Printing the fitted model to predict for generalized linear model

According to Figure 15, the predicted history of smoking score for a recurrence in patients with differentiated thyroid cancer where the patient is a 55-year-old female who has a history of smoking, with N1b, T1b, M0, a subclinical hypothyroidism, and is in stage II. Therefore, giving a 55.6% based on our computation, where $Hxsmoker^0 = 0.3665 + 0.0018 \cdot Age\ 55 - 0.1411 \cdot Female + 0.0401\ Yes\ History\ of\ Smoking -$

$$0.1526 \cdot \text{Subclinical Hypothyroidism} + 0.0849 \cdot T1b + 0.2414 \cdot N1b - 0.0868 \cdot M0 + 0.1046 \cdot \text{Stage II} = 0.556\%.$$

Results using Binary Logistic Regression Model with a binomial logit link function

In this result, the data cleaning was applied using the Binary Logistic Regression Model with the binomial logit link function, as mentioned above. MASS was needed to install packages() in the R programming language. MASS support functions and datasets for Venables and Ripley's MASS [41]. Package MASS adds methods for glm and nls fits [42]. In addition, we will create indicator variables to represent the categorical variables in the binary logistic regression model. By doing so, we fit the generalized linear model and then print a summary of our model, which is shown in Figure 15.

```
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
     response.rel, family = binomial(link = logit), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      1.32041    1.68996   0.781 0.434613
age              0.04856    0.02183   2.225 0.026078 *
gender.relF      -0.92738    0.72338  -1.282 0.199843
hxsmoking.relNo  -0.20705    1.36935  -0.151 0.879814
n.relN1a         2.28599    0.87184   2.622 0.008741 **
n.relN1b         2.54797    0.78142   3.261 0.001111 **
response.relBiochemical Incomplete -3.33817    0.94387  -3.537 0.000405 ***
response.relExcellent -8.13436    1.27405  -6.385 1.72e-10 ***
response.relIndeterminate -5.94255    0.97037  -6.124 9.13e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 442.657 on 363 degrees of freedom
Residual deviance: 84.742 on 355 degrees of freedom
AIC: 102.74

Number of Fisher Scoring iterations: 8
```

Figure 16. Summary of Binary Logistic Regression Model with a binomial logit link function

Figure 16 displays a summary of the Binary Logistic Regression Model with a binomial logit link function. Figure 14, shows a fitted generalized linear model with the glm() function being utilized. In this binary logistic regression model, only a few

indicator variables were used, such as age, gender, hxsmoking, n, and response. After doing so, we will specify all the estimated parameters. From this output, the fitted binary logistic regression model is: $\ln \frac{\hat{p}(treatment)}{1-\hat{p}(treatment)} = 1.3204 + 0.0486 \cdot Age - 0.9274 \cdot Female - 0.2070 \cdot No\ History\ of\ Smoking + 2.2860 \cdot N1a + 2.5480 \cdot N1b - 3.3382 \cdot Biochemical\ Incomplete\ Response - 8.1344 \cdot Excellent\ Response - 5.9425 \cdot Indeterminate\ Response$. Since the p-value ($1.849879e-72$) is small and not more than 0.05, we conclude that the fitted model fits better than the intercept-only model. Age, N1a, N1b, Biochemical Incomplete, Excellent, and Indeterminate are statistically significant predictors of differentiated thyroid cancer recurrence score at the 5% significance level, since the corresponding values are less than 0.05. As age increases by one year, the estimated odds increase by $(\exp\{0.0486\} - 1) \cdot 100\% = 1.04980034661 - 1 = 0.0498003466 = 0.0498003466 \cdot 100 = 4.98\%$ for a differentiated thyroid cancer recurrence patient. In this Binary Logistic Regression model, the estimated odds in favor of treatment from differentiated thyroid cancer recurrence for patients with N1a increased $\exp\{2.2860\} \cdot 100\% = 983.5517 = 983.55\%$ then those of patients without N1a. For N1b, the estimated odds in favor of treatment from differentiated thyroid cancer recurrence for patients with N1b increased $\exp\{2.5480\} \cdot 100\% = 1278.152 = 1278.15\%$ then those of patients without N1b. Moreover, for T, the odds of recurrence for a patient with a Biochemical Incomplete response is $\exp\{-3.3382\} \cdot 100\% = 0.03550080169 \cdot 100\% = 3.550080169 = 3.55\%$; for the odds of recurrence for a patient with an Excellent response is $\exp\{-8.1344\} \cdot 100\% = 2.93274948 \cdot 10^{-4} \cdot 100\% = 0.0293274948 = 0.03\%$, and

for an Indeterminate response, the estimated probability is $\exp\{-5.9425\} \cdot 100\% = 0.002625457795 \cdot 100\% = 0.2625457795 = 0.26\%$.

Supposed that we wanted to find the predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response using the Binary Logistic Regression Model with binomial logit link function. First let us compute the predicted binary logistic regression model as:

$$P^0(\text{treatment}) = \frac{1}{-1.3204 - 0.0486 \cdot \text{Age } 50 + 0.9274 \cdot \text{Female} - 2.5480 \cdot \text{N1b} + 5.9425 \cdot \text{Indeterminate Response}}$$

$$= \frac{1}{2.770904} = 0.3608931 \approx 0.3609 \approx 36.09\%$$

```
binary.prediction <- predict(binary.fitted.model, data.frame(age=50, gender.rel="F",
  hxsmoking.rel="Yes", n.rel="N1b", response.rel = "Indeterminate"), type="response")
print(binary.prediction)
```

1
0.3604712

Figure 17. Printing the fitted model to predict for binary logistic regression

Based to Figure 17, the predicted history of treatment score for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response is $0.36 \approx 36\%$.

Results using Probit Model with a binomial probit link function

The same approach was applied to these results, with data cleaning and factor and releval functions to manage categorical variables. From this, we obtain a summary, as depicted in Figure 18.

```

glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
     response.rel, family = binomial(link = "probit"), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.80461    0.84237   0.955 0.339491
age            0.02310    0.01063   2.172 0.029828 *
gender.relF    -0.49739    0.36834  -1.350 0.176902
hxsmoking.relNo -0.14012    0.65086  -0.215 0.829546
n.relN1a       0.96566    0.45397   2.127 0.033408 *
n.relN1b       1.34384    0.37236   3.609 0.000307 ***
response.relBiochemical Incomplete -1.70501    0.45442  -3.752 0.000175 ***
response.relExcellent -4.14319    0.53172  -7.792 6.59e-15 ***
response.relIndeterminate -3.06370    0.42804  -7.158 8.21e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 442.66  on 363  degrees of freedom
Residual deviance:  85.82  on 355  degrees of freedom
AIC: 103.82

Number of Fisher Scoring iterations: 9

```

Figure 18. Summary of Probit Model with a binomial logit link function

Figure 18 portrays a summary of the Probit Model with a binomial logit link function. Figure 18 shows a fitted generalized linear model with the `glm()` function employed, as in the Binary Logistic Regression Model. The same indicator variables used in the Binary Logistic Regression Model are utilized in the Probit Model. Therefore, from this output, the fitted Probit model is: $\varphi^{-1}(\hat{P}(\text{treatment})) = 0.8046 + 0.0231 \cdot \text{Age} - 0.4974 \cdot \text{Female} - 0.1401 \cdot \text{No History of Smoking} + 0.9657 \cdot \text{N1a} + 1.3438 \cdot \text{N1b} - 1.7050 \cdot \text{Biochemical Incomplete Response} - 4.1432 \cdot \text{Excellent Response} - 3.0637 \cdot \text{Intermediate Response}$. Since the p-value is small and not more than 0.05, the fitted model fits better than the intercept-only model. In the Binary Logistic Regression and Probit Model, we detected that the p-value was small. At the 5% significance level, age, N1a, N1b, biochemical incomplete, excellent, and indeterminate are significant predictors, as in the Probit model. The z-score for the estimated probability of treatment from differentiated thyroid cancer recurrences for patients with Biochemical Incomplete Response is decreased by -1.7050 than that for

patients with Biochemical Incomplete Response. Moreover, for patients with an Excellent response, it is decreased larger by -4.1432 than that for patients with an Excellent response. Last, for patients with Indeterminate response, it is decreased by -3.0637, then that for patients with Indeterminate response.

Supposed that we wanted to find the predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response using the Probit Model with binomial probit link function. The predicted probit model is obtained as: $P^0(treatment) = \varphi(0.8046 + 0.0231 \cdot Age\ 50 - 0.4974 \cdot Female + 1.3438 \cdot N1b - 3.0637 \cdot Intermediate\ Response) = \varphi(-0.2577) = 0.3983192 \approx 0.3983 \approx 39.83\%$. This predicted value is slightly smaller than that in the binary logistic regression model. Moreover, in Figure 19, we will display the printing the fitted model to predict for probit model which is shown below.

```
probit.prediction <- predict(probit.fitted.model,
data.frame(age=50, gender.rel="F", hxsmoking.rel="Yes",
n.rel="N1b", response.rel = "Indeterminate"), type="response")
print(probit.prediction)

      1
0.3982731

probit = (0.8046+0.0231*50-0.4974+1.3438-3.0637)
print(probit)
print(pnorm(probit))

[1] -0.2577
[1] 0.3983192
```

Figure 19. Printing the fitted model to predict for probit model

Results using Complementary Log-Log Model with a binomial cloglog link function

The Complementary Log-Log Model uses the same approach as the Probit and Binary Logistic Regression Model that was applied to these results, with data cleaning and factor and relevel functions to manage categorical variables. After running the data cleaning and factor and relevel functions to manage categorical variables, we display a summary of our Complementary Log-Log Model shown in Figure 20.

```
Call:
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
     response.rel, family = binomial(link = "cloglog"), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.60498    0.91218   0.663 0.507146
age            0.02040    0.01196   1.705 0.088242 .
gender.relF    -0.98466    0.47062  -2.092 0.036415 *
hxsmoking.relNo -0.06733    0.64150  -0.105 0.916407
n.relN1a       0.78447    0.53236   1.474 0.140593
n.relN1b       1.68642    0.50957   3.309 0.000935 ***
response.relBiochemical Incomplete -1.35855    0.43955  -3.091 0.001996 **
response.relExcellent -6.00962    1.04324  -5.761 8.38e-09 ***
response.relIndeterminate -3.49937    0.53919  -6.490 8.58e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 442.657  on 363  degrees of freedom
Residual deviance:  87.432  on 355  degrees of freedom
AIC: 105.43

Number of Fisher Scoring iterations: 10
```

Figure 20. Summary of Complementary Log-Log Model with a binomial cloglog link function

Figure 20 portrays a summary of the Complementary Log-Log Model with a binomial cloglog link function, just like in the Probit Model with a binomial logit link function and the Binary Logistic Regression Model with a binomial logit link function. Also, a fitted generalized linear model is shown in Figure 19, as we can see with family = binomial(link = "cloglog") and the data and the coefficients shown with AIC: 105.43. The same indicator variables used in the Binary Logistic Regression and Probit model are

employed in the Complementary Log-Log model. Consequently, from this output, the fitted Complementary Log-Log model is:

$$1 - \hat{P}(treatment) = \exp(-\exp(0.6050 + 0.0204 \cdot Age - 0.9847 \cdot Female - 0.0673 \cdot No\ History\ of\ Smoking + 0.7845 \cdot N1a + 1.6864 \cdot N1b - 1.3585 \cdot Biochemical\ Incomplete\ Response - 6.0096 \cdot Excellent\ Response - 3.4994 \cdot Indeterminate\ Response)).$$

Since the p-value is small and not more than 0.05, the fitted model fits better than the intercept-only model. In the Binary Logistic Regression and Probit Model, we detected that the p-value was small. The age, N1a, N1b, biochemical incomplete, excellent, and indeterminate, is still a significant predictor at the 5% level, since their p-values are smaller than 0.05, as seen in the Binary Logistic Regression and Probit model. As age increases by one, the estimated probability is raised to the power $\exp\{0.0204\} = 1.020609502 = 1.0206$. Patients with N1b, where the cancer has spread to other lymph nodes in the neck is raised to the power $\exp\{1.6864\} = 5.400005651 = 5.4000$. However, with patients in biochemical incomplete response, decreases is raised to the power $\exp\{-1.3585\} = 0.257046057 = 0.2570$, and excellent response where the patient has no clinical, biochemical, or structural evidence of disease is present after treatment is raised to the power $\exp\{-6.0092\} = 0.002456052236 = 0.0024$, and last the patient with an indeterminate response where the patient has a response non-specific biochemical or structural findings which cannot be confidently classified as either benign or malignant is raised to the power $\exp\{-3.994\} = 0.01842586306 = 0.0184$.

Supposed that we wanted to find the predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of

smoking, N1b, and an indeterminate response using the Complementary Log-Log model with a binomial cloglog link function. The predicted complementary log-log model is derived as: $P^0(treatment) = 1 - \exp(-\exp(0.6050 + 0.0204 \cdot Age\ 50 - 0.9847 \cdot Female + 1.6864 \cdot N1b - 3.4994 \cdot Indeterminate\ Response)) = 0.2662083 \approx 0.2662 \approx 26.62\%$. This predicted value is slightly smaller than that in the binary logistic regression model and probit. In other words, in Figure 21, we will show the printing of the fitted model to predict the Complementary Log-Log model, which is indicated below.

```
cloglog.prediction <- predict(cloglog.fitted.model, data.frame(age=50,
gender.rel="F", hxsmoking.rel="Yes", n.rel="N1b",
response.rel = "Indeterminate"), type="response")
print(cloglog.prediction)

      1
0.2661779

cloglog=1-exp(-exp(0.6050+0.0204*50-0.9847+1.6864-3.4994))
print(cloglog)

[1] 0.2662083
```

Figure 21. Printing the fitted model to predict for Complementary Log-Log model

Table 3

AIC (Akaike Information Criterion), AICc (corrected AIC), and
BIC (Bayesian Information Criterion) Values

	Binary Logistic Regression	Probit Model	Complementary Log-Log
AIC	102.7418	103.8201	105.432
AICc	103.2503	104.3286	105.9404
BIC	137.8162	138.8945	140.5064

Thus, to compare the fitted logistic, probit, and complementary log-log models in terms of goodness-of-fit, we compare the AIC, AICc, and BIC criteria values. The AIC (Akaike Information Criterion) was introduced by Hirotugu Akaike in 1973, and the BIC (Bayesian Information Criterion) by Gottfried Schwarz in 1978, while the AICc (corrected AIC) was developed by Tadashi Sugiura in 1978. Lastly, in Table 3, the Binary Logistic Regression with 102.7418 for AIC, 103.2503 for AICc, and BIC 137.8162 has the smallest values in all AIC, AICc, and BIC criteria. We conclude that the Binary Logistic Regression model should be chosen over the other Probit and Complementary Log-Log (cloglog) models. Therefore, the Binary Logistic Regression model has the best fit.

Results using Random Forest

The Random Forest uses a machine learning algorithm. Data cleaning and factor functions are employed like in the Generalized Linear Model, Binary Logistic Regression, Probit, and Complementary Log-Log Model. After all, have been operated, we build a regression tree on the training set with the RSS splitting criterion to model Differentiated Thyroid Cancer Recurrence. In the Random Forest, we split the data into 70% training and 30% for testing; therefore, we balanced out the sets because our sets needed to be equal to 100%. By adding both 70 and 30, we obtain 100. So, this is good practice for training and testing our data. The seed was set to 123 in R. Setting the value to 123 in R is to operate a random number generator's seed. Replace was set to true, and a boolean array with a probability of 0.7 for True and 0.3 for False was created. Next, we train a Random Forest Regressor to model Differentiated Thyroid Cancer Recurrence. In

the training set, recurred was the target variable, and all other variables were used, such as age, gender, smoking, hxsmoking, hxradiation, thyroid function, physical examination, adenopathy, pathology, focality, risk, t, n, m, stage, and response were trained along with the data, number of trees to be fitted [43] to 100, number of variable to be sampled as split criteria at each node [43] was set to 5, the maximum number of terminal nodes [43] to 30, proximity to true, importance to true.

Cell Contents			
		N	
Chi-square contribution			
N / Row Total			
N / Col Total			
N / Table Total			

Total Observations in Table: 261			
----------------------------------	--	--	--

diff_thyroid_cancer_recurrence_preds_train	train\$recurred		Row Total
	No	Yes	
No	184	1	185
	22.011	52.597	
	0.995	0.005	0.709
	1.000	0.013	
	0.705	0.004	
Yes	0	76	76
	53.579	128.032	
	0.000	1.000	0.291
	0.000	0.987	
	0.000	0.291	
Column Total	184	77	261
	0.705	0.295	

Figure 22. Crosstable of the Random Forest

After the training set, we computed the crosstable calculations in Figure 22 for the training stage. A Crosstable is a package centered on a single function, `crosstable()`,

which easily computes descriptive statistics on datasets [44]. In Figure 22, in the Crosstable of the Random Forest, we see that when the training set was split into 70% and 30%, we know how the 'Yes' and 'No' are depicted in the table along with the row total and the column total. Hence, in Row Total, we get 185. This is because we add $184 + 1 = 185$; for 76, we add $76 + 0 = 76$. While a Confusion Matrix is depicted in Figure 23. Generally, a Crosstable is the same as a Confusion Matrix. We compute the confusion matrix to evaluate the accuracy of a classification [45]. Thus, in binary classification, the count of true negatives is $C_{0,0}$, false negatives is $C_{1,0}$, true positives is $C_{1,1}$, and false positives is $C_{0,1}$ [45]. True Positive (TP) correctly predicts yes, False Negative (FN) incorrectly predicts yes, True Negative (TN) incorrectly predicts no, and False Positive (FP) incorrectly predicts no. Figure 23 shows a Confusion Matrix where the True Positive is 76, the False Positive is 1, the True Negative is 184, and the False Negative is 0; see below. It also shows a value where the highest is above 150; therefore, the highest value is True Negative, whereas the lowest value, 0, is False Negative. Based on the Confusion Matrix, we can obtain the accuracy, precision, recall, and F1 score. Because, we already got the True Positive, False Positive, True Negative, False, Negative, now we use need the Total Numbers of TP, FP, TN, and FN, by doing this we obtain 261.

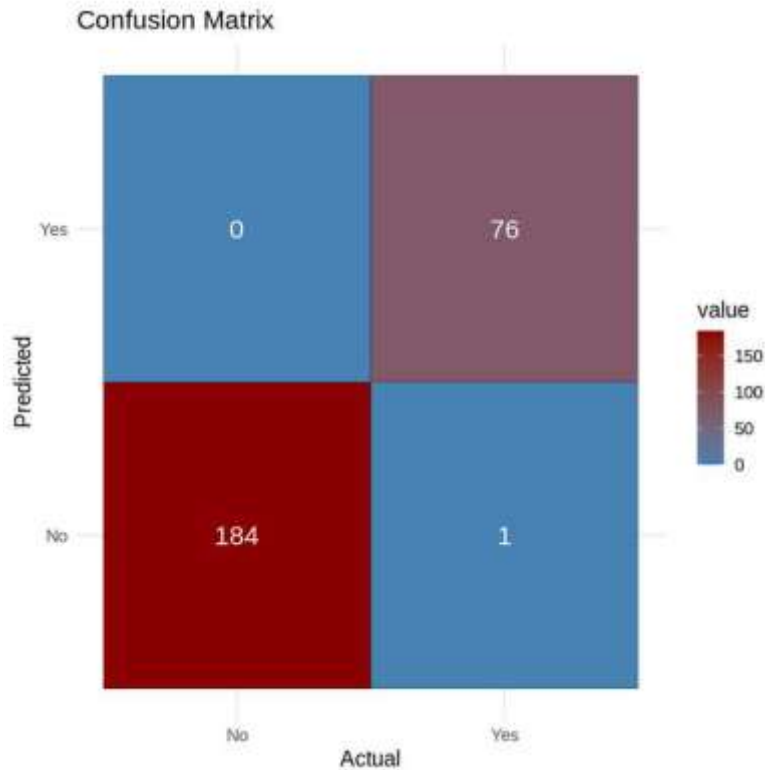


Figure 23. Confusion Matrix of the Random Forest

```

Accuracy:          0.9962
Misclass. Rate:    0.0038
Precision:         0.987
Recall (Sensitivity): 1
Specificity:       0.9946
F1 Score:          0.9935
False Positive Rate (FPR): 0.0054
False Negative Rate (FNR): 0
True Positive Rate (TPR): 0.4108
True Negative Rate (TNR): 2.4211
Negative Predictive Value (NPV): 1
One Minus Specificity: 0.0054

```

Figure 24. Accuracy, Precision, Recall, and F1 Score of the Random Forest

Figure 24, exhibits the accuracy with 99.62%, precision with 98.7%, recall with 1%, and F1 score 99.35%. This is how we do the arithmetic in accuracy where True Positive + True Negative / Total Numbers of TP, FP, TN, and FN, is $76 + 184 / 261 =$

$260 / 261 = 99.62\%$. In Precision, Precision is $\text{True Positive} / \text{True Positive} + \text{False Positive} = 76 / 76 + 1 = 76 / 77 = 98.70\%$. In Recall score, $\text{True Positive} / \text{True Positive} + \text{False Negative} = 76 / 76 + 0 = 76 / 76 = 1.0\%$. And the F1 Score, $2 * \text{Precision} * \text{Recall} / \text{Precision} + \text{Recall} = 2 * 0.987 * 1.0 / 0.987 + 1.0 = 1.974 / 1.987 = 99.35\%$. Lastly, we will use the accuracy vs the Gini score of the Random Forest to analyze the variable importance plot. In Figure 25 – see below, we see two plots: the accuracy, which is on the left-hand side of the image, and the Gini, which is on the right-hand side. Computing variable importance (VI) and communicating them through variable importance plots (VIPs) is a fundamental component of IML [46]. Based on Figure 25, response accounts for both high accuracy and Gini. Meanwhile, pathology accounts for low accuracy and Gini. Mean Decrease Accuracy (MDA) and Mean Decrease Gini (MDG) were operated because we wanted to see how much the model's accuracy decreased in both accuracy and Gini. Therefore, this method was applied.

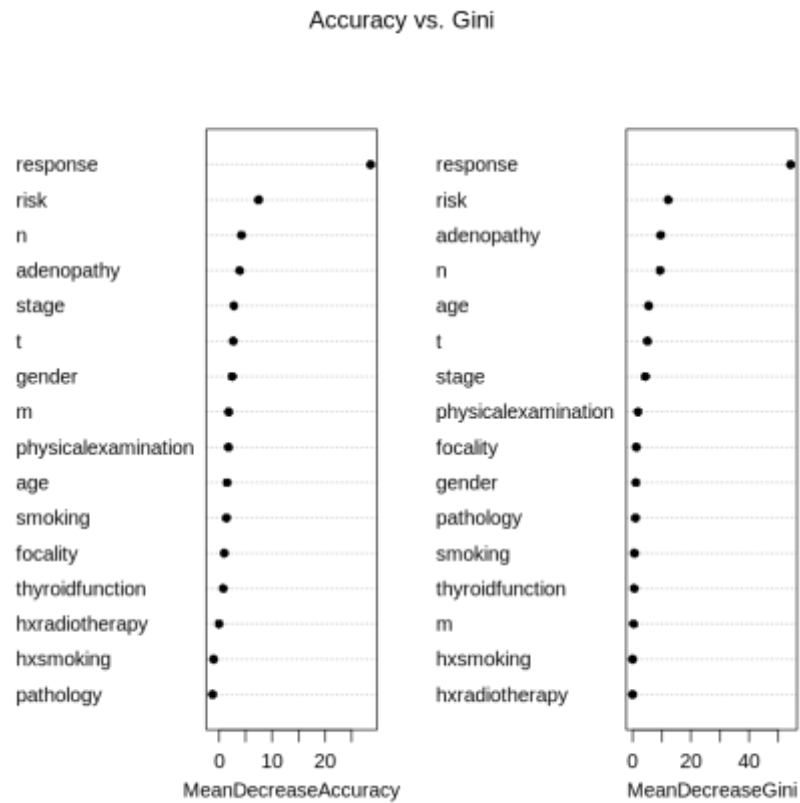


Figure 25. Accuracy vs. Gini Score of the Random Forest

CHAPTER 4

Conclusion

Chapter 4 will finalize the conclusion, the results, and future work.

In the Differentiated Thyroid Cancer Recurrence dataset, the Random Forest, performed the best with 99.62% accuracy, while the Generalized Linear Model had an accuracy of 55.6%, the Binary Logistic Regression Model had an accuracy of 36.04%, the Probit Model had an accuracy of 39.83%, the Complementary Log-Log Model had an accuracy of 26.62% which is the lowest accuracy.

Our future work will be to test and train more Machine Learning algorithms for the Differentiated Thyroid Cancer Recurrence dataset, specifically neural networks. Moreover, we plan to utilize Convolution Neural Networks and Artificial Neural Networks to test and train our model with machine learning algorithms. Also, we plan to find any Machine Learning techniques to reduce overfitting and improve accuracy in the Differentiated Thyroid Cancer Recurrence dataset. Lastly, we will utilize multi-class classification for a single dataset to train and test the data.

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